

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(a)    | <p><i>any <b>two</b> from:</i><br/> <i>ref. to</i> can organise / order, enzymes / events, in pathway ;</p> <p>consequence in terms of distance ;<br/> e.g. product of one reaction is close to next enzyme (where it acts as substrate)<br/> enzyme (more likely to be) closer to substrate<br/> distance between enzymes shorter<br/> I short distance / less distance to travel, unqualified</p> <p>(so) increases chance of successful collisions between substrate and enzymes ;</p> <p>increased rate of formation of final product<br/> <b>or</b><br/> decrease time for ES complex to occur ;</p> <p><i>idea that</i> products can pass to either side of the membrane ;</p> <p>all reactions, localised / occur in the same location (within the cell) ;</p> <p>AVP ; e.g. membrane provides a source of arachidonic acid<br/> I phospholipid</p> | <b>2</b> |
| 5(b)(i) | <p>(similar shaped) curve to the right of main curve <u>and</u> reaches <math>V_{\max}</math> ;<br/> <b>A</b> increasing to reach <math>V_{\max}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>1</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(b)(ii) | stays the same ;<br><br>increases ;                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
| 5(c)(i)  | the gene / PTGS2, is a sequence of nucleotides, that forms part of a DNA molecule / AW / on chromosome 1 ;<br><b>A</b> bases<br><br>codes for (the production of) a, polypeptide / enzyme / COX-2 ;<br><b>A</b> protein <b>I</b> codes for an amino acid<br><br><i>max one if no example used</i>                                                                                                                                                  | 2     |
| 5(c)(ii) | <i>any <b>two</b> from:</i><br>more, product / compounds, produced that stimulate mitosis ;<br><br>increased, DNA replication / cell division / mitosis ;<br><b>I</b> <i>ref. to</i> cancer cells replicating<br><br><i>idea that</i> , may lead to / increases chance of / AW, (other) mutations ;<br><br>mutations can result in oncogenes ;<br><br>AVP ; e.g.<br><i>idea that</i> proof checking capacity impaired / increased chance of errors | 2     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks    |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(d)     | <p><i>any <b>three</b> from:</i></p> <p>phospholipids may be, used in / added to, the (cell surface) membrane ;</p> <p>arachidonic acid, is unsaturated / polyunsaturated ;</p> <p>has, C=C / (carbon-carbon) double bonds ;<br/> <i>allow ecf if stated as saturated</i></p> <p>unsaturated fatty acid tails, have kinks / not linear / AW ;<br/> <b>A</b> double bonds produce, kinks / AW</p> <p>increased distance between, phospholipids / other fatty acid tails ;<br/> <b>A</b> phospholipids cannot pack closely together</p> <p>less hydrophobic interactions between phospholipid (molecules) ;</p> | <b>3</b> |

| Question | Answer | Marks |
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